Assignment No: 3

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Code:

```
import pandas as pd
from scipy.stats import ttest_ind, chi2_contingency

# Load Dataset
df = pd.read_excel("Student_Performance_Dataset.xlsx")

#A] Independent Samples t-Test

# Compare Math Scores of Male and Female Students
# Extract math scores

male_scores = df[df["Gender"] == "Male"]["Math_Score"]
female_scores = df[df["Gender"] == "Female"]["Math_Score"]

# Perform t-test

t_stat, p_value = ttest_ind(
    male_scores,
    female_scores,
    equal_var=False      # Welch's t-test
)

#performs an Independent
# Two-Sample t-test (Welch's t-test) to compare the average mathematics scores of male and
female students.

print("=" * 60)
print("INDEPENDENT SAMPLE t-TEST")
print("=" * 60)

print("Mean Math Score (Male) :", round(male_scores.mean(), 2))
print("Mean Math Score (Female) :", round(female_scores.mean(), 2))

print("\nT-statistic :", round(t_stat, 4))
print("P-value     :", round(p_value,
4))

alpha = 0.05
```

```
if p_value < alpha:
    print("\nResult: Reject the Null Hypothesis")
    print("There is a significant difference in mathematics scores.")
else:
    print("\nResult: Fail to Reject the Null Hypothesis")
    print("No significant difference in mathematics scores.")

    #H0:Null Hypothesis:No significant difference in mathematics scores
    #H1:significant difference in mathematics scores
```

Output:

```
============================================================
INDEPENDENT SAMPLE t-TEST
============================================================
Mean Math Score (Male)   : 67.72
Mean Math Score (Female) : 67.78

T-statistic : -0.1148
P-value     : 0.9086

Result: Fail to Reject the Null Hypothesis
No significant difference in mathematics scores.
```

```
# B]Chi-Square Test
# Relationship between Parental Education and Performance
print("\n" + "=" * 60)
print("CHI-SQUARE TEST OF INDEPENDENCE")
print("=" * 60)
# Create contingency table
contingency_table = pd.crosstab(
    df["Parental_Education"],
    df["Performance_Level"]
)

print("\nContingency Table:\n")
print(contingency_table)

# Perform Chi-square test
```

```
chi2, p, dof, expected = chi2_contingency(contingency_table)

print("\nChi-Square Statistic :", round(chi2, 4))
print("Degrees of Freedom :", dof)
print("P-value            :", round(p, 4))

if p < alpha:
    print("\nResult: Reject the Null Hypothesis")
    print("Parental education and student performance are significantly associated.")
else:
    print("\nResult: Fail to Reject the Null Hypothesis")
    print("No significant association exists between parental education and student
performance.")

print("\nExpected
Frequencies:\n")
print(pd.DataFrame(
    expected,
    index=contingency_table.index,
    columns=contingency_table.columns
))
```

Output:

```
============================================================
CHI-SQUARE TEST OF INDEPENDENCE
============================================================

Contingency Table:

Performance_Level   Average  Excellent  Good  Poor
Parental_Education
Graduate                648        184   744    85
High School             642        203   730    85
Postgraduate            661        206   717    95

Chi-Square Statistic : 2.8443
Degrees of Freedom   : 6
P-value              : 0.8281

Result: Fail to Reject the Null Hypothesis
No significant association exists between parental education and student performance.

Expected Frequencies:

Performance_Level    Average  Excellent      Good    Poor
Parental_Education
Graduate            648.1222   196.9946  727.8502  88.033
High School         647.7320   196.8760  727.4120  87.980
Postgraduate        655.1458   199.1294  735.7378  88.987
```